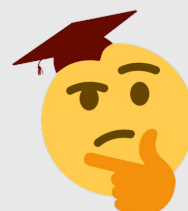


IB Diploma

MATHEMATICS AA

HL P1
2012—2022

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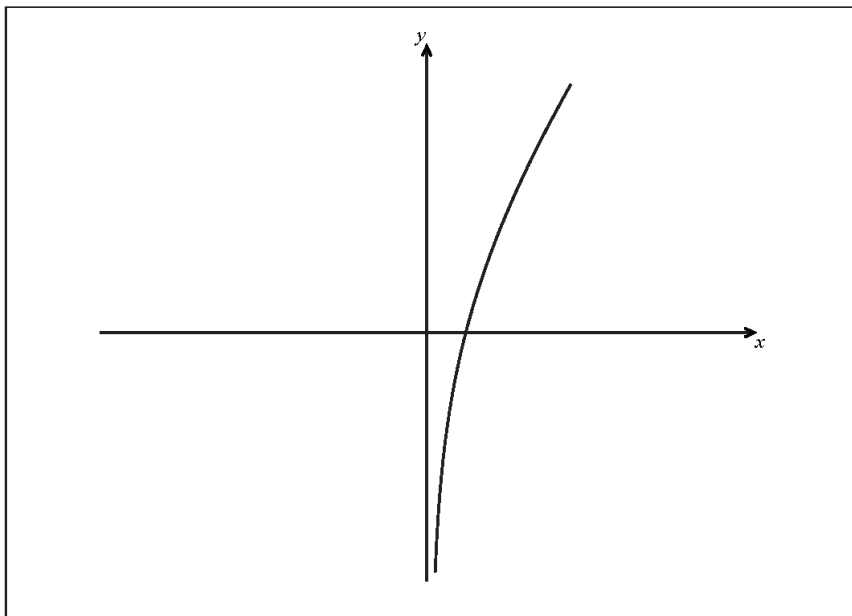


1-(MAT/11_HL_Summer_2012_Q4)-Graphs, Functions-Roots

The graph below shows $y = f(x)$, where $f(x) = x + \ln x$.

- (a) On the graph below, sketch the curve $y = f^{-1}(x)$.

[2 marks]



- (b) Find the coordinates of the point of intersection of the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$.

[4 marks]

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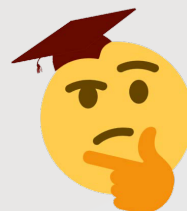
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2-(MAT/12_HL_Summer_2012_Q10)-Functions-Roots, Graphs, Integration

The function f is defined on the domain $\left[0, \frac{3\pi}{2}\right]$ by $f(x) = e^{-x} \cos x$.

(a) State the two zeros of f .

[1 mark]

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(b) Sketch the graph of f .

[1 mark]



- (c) The region bounded by the graph, the x -axis and the y -axis is denoted by A and the region bounded by the graph and the x -axis is denoted by B . Show that the ratio of the area of A to the area of B is

$$\frac{e^{\pi} \left(e^{\frac{\pi}{2}} + 1 \right)}{e^{\pi} + 1},$$

[7 marks]

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There is a vertical margin line on the left side, creating a narrow left margin. The paper appears to be from a notebook or a standard ruled sheet of paper.

3-(MAT/12_HL_Summer_2012_Q11)-Functions-Roots,Probability

Consider the following functions:

$$f(x) = \frac{2x^2 + 3}{75}, x \geq 0$$

$$g(x) = \frac{|3x - 4|}{10}, x \in \mathbb{R}.$$

- (a) State the range of f and of g . [2 marks]

- (b) Find an expression for the composite function $f \circ g(x)$ in the form $\frac{ax^2 + bx + c}{3750}$, where a, b and $c \in \mathbb{Z}$. [4 marks]

- (c) (i) Find an expression for the inverse function $f^{-1}(x)$.

- (ii) State the domain and range of f^{-1} . [4 marks]

The domains of f and g are now restricted to $\{0, 1, 2, 3, 4\}$.

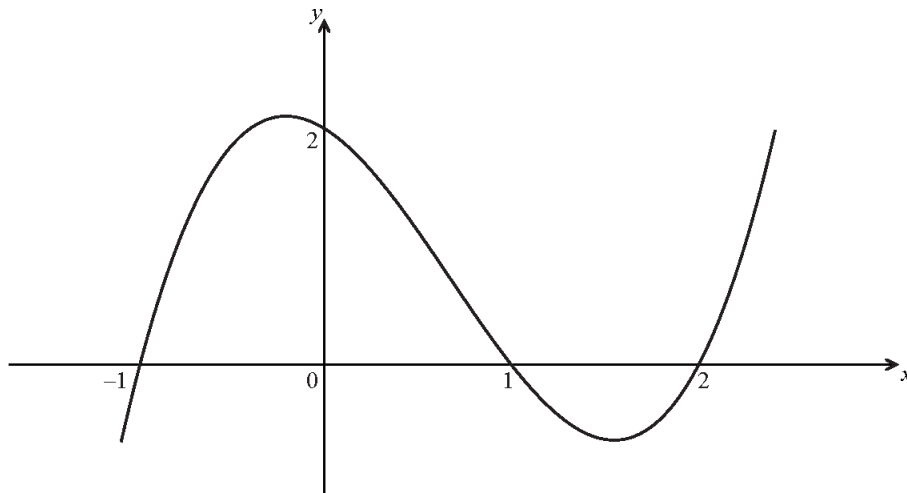
- (d) By considering the values of f and g on this new domain, determine which of f and g could be used to find a probability distribution for a discrete random variable X , stating your reasons clearly. [6 marks]

- (e) Using this probability distribution, calculate the mean of X . [2 marks]



4-(MAT/10_HL_Winter_2012_Q3)-Functions-Roots, Graphs

Let $f(x) = x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{Z}$. The diagram shows the graph of $y = f(x)$.



- (a) Using the information shown in the diagram, find the values of a , b and c . [4 marks]

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(b) If $g(x) = 3f(x-2)$,

- (i) state the coordinates of the points where the graph of g intercepts the x -axis.

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- (ii) Find the y -intercept of the graph of g .

[3 marks]

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5-(MAT/10_HL_Winter_2012_Q12)-Functions-Roots,MathematicalInduction

Consider a function f , defined by $f(x) = \frac{x}{2-x}$ for $0 \leq x \leq 1$.

- (a) Find an expression for $(f \circ f)(x)$. [3 marks]

Let $F_n(x) = \frac{x}{2^n - (2^n - 1)x}$, where $0 \leq x \leq 1$.

- (b) Use mathematical induction to show that for any $n \in \mathbb{Z}^+$

$$\underbrace{(f \circ f \circ \dots \circ f)}_{n \text{ times}}(x) = F_n(x). \quad [8 \text{ marks}]$$

- (c) Show that $F_n(x)$ is an expression for the inverse of F_n . [6 marks]

- (d) (i) State $F_n(0)$ and $F_n(1)$.

- (ii) Show that $F_n(x) < x$, given $0 < x < 1$, $n \in \mathbb{Z}^+$.

- (iii) For $n \in \mathbb{Z}^+$, let A_n be the area of the region enclosed by the graph of F_n^{-1} , the x -axis and the line $x=1$. Find the area B_n of the region enclosed by F_n and F_n^{-1} in terms of A_n . [6 marks]



6-(MAT/12_HL_Summer_2013_Q12)-Functions-Roots, Graphs

The function f is defined by $f(x) = \frac{2x-1}{x+2}$, with domain $D = \{x : -1 \leq x \leq 8\}$.

(a) Express $f(x)$ in the form $A + \frac{B}{x+2}$, where A and $B \in \mathbb{Z}$. [2 marks]

(b) Hence show that $f'(x) > 0$ on D . [2 marks]

(c) State the range of f . [2 marks]

(d) (i) Find an expression for $f^{-1}(x)$.

(ii) Sketch the graph of $y = f(x)$, showing the points of intersection with both axes.

(iii) On the same diagram, sketch the graph of $y = f^{-1}(x)$. [8 marks]

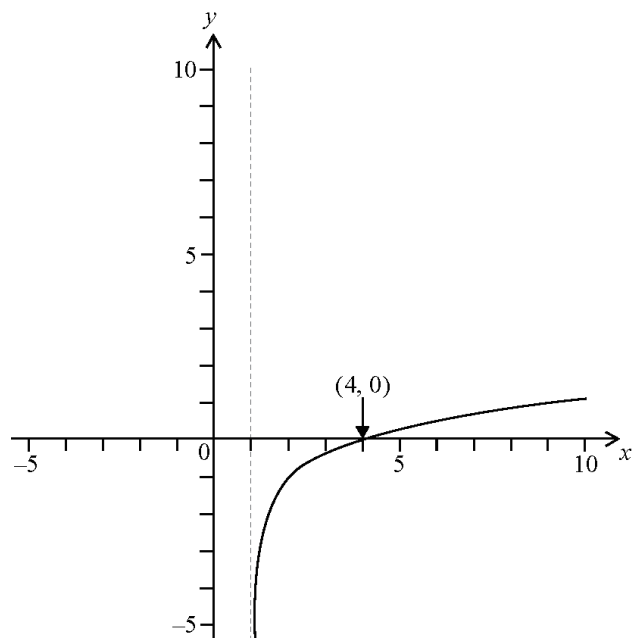
(e) (i) On a different diagram, sketch the graph of $y = f(|x|)$ where $x \in D$.

(ii) Find all solutions of the equation $f(|x|) = -\frac{1}{4}$. [7 marks]



7-(MAT/10_HL_Winter_2013_Q3)-Functions-Roots, Graphs

The diagram below shows a sketch of the graph of $y = f(x)$.



- (a) Sketch the graph of $y = f^{-1}(x)$ on the same axes. [2]
- (b) State the range of f^{-1} . [1]
- (c) Given that $f(x) = \ln(ax + b)$, $x > 1$, find the value of a and the value of b . [4]



8-(MAT/11_HL_Summer_2014_Q4)-Functions-Roots

[Maximum mark: 6]

The equation $5x^3 + 48x^2 + 100x + 2 = a$ has roots r_1 , r_2 and r_3 .
Given that $r_1 + r_2 + r_3 + r_1 r_2 r_3 = 0$, find the value of a .

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9-(MAT/12_HL_Summer_2014_Q8)-Functions-Roots

The function f is defined by

$$f(x) = \begin{cases} 1 - 2x, & x \leq 2 \\ \frac{3}{4}(x-2)^2 - 3, & x > 2 \end{cases}$$

(a) Determine whether or not f is continuous.

[2]

The graph of the function g is obtained by applying the following transformations to the graph of f :

a reflection in the y -axis followed by a translation by the vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

(b) Find $g(x)$.

[4]

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10-(MAT/12_HL_Summer_2014_Q14)-Functions-Roots

Consider the following functions:

$$h(x) = \arctan(x), \quad x \in \mathbb{R}$$

$$g(x) = \frac{1}{x}, \quad x \in \mathbb{R}, \quad x \neq 0$$

(a) Sketch the graph of $y = h(x)$. [2]

(b) Find an expression for the composite function $h \circ g(x)$ and state its domain. [2]

Given that $f(x) = h(x) + h \circ g(x)$,

(c) (i) find $f'(x)$ in simplified form;

(ii) show that $f(x) = \frac{\pi}{2}$ for $x > 0$. [7]

Nigel states that f is an odd function and Tom argues that f is an even function.

(d) (i) State who is correct and justify your answer.

(ii) Hence find the value of $f(x)$ for $x < 0$. [3]



11-(MAT/10_HL_Winter_2014_Q1)-Functions-Roots, Graphs

The function f is defined by $f(x) = \frac{1}{x}$, $x \neq 0$.

The graph of the function $y = g(x)$ is obtained by applying the following transformations to the graph of $y = f(x)$:

a translation by the vector $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$;

a translation by the vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

(a) Find an expression for $g(x)$. [2]

(b) State the equations of the asymptotes of the graph of g . [2]

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12-(MAT/10_HL_Winter_2014_Q11)-Functions-Roots,Differentiation,Integration

The function f is defined as $f(x) = e^{3x+1}$, $x \in \mathbb{R}$.

(a) (i) Find $f^{-1}(x)$.

(ii) State the domain of f^{-1} . [4]

The function g is defined as $g(x) = \ln x$, $x \in \mathbb{R}^+$.

The graph of $y = g(x)$ and the graph of $y = f^{-1}(x)$ intersect at the point P.

(b) Find the coordinates of P. [5]

The graph of $y = g(x)$ intersects the x -axis at the point Q.

(c) Show that the equation of the tangent T to the graph of $y = g(x)$ at the point Q is $y = x - 1$. [3]

A region R is bounded by the graphs of $y = g(x)$, the tangent T and the line $x = e$.

(d) Find the area of the region R . [5]

(e) (i) Show that $g(x) \leq x - 1$, $x \in \mathbb{R}^+$.

(ii) By replacing x with $\frac{1}{x}$ in part (e)(i), show that $\frac{x-1}{x} \leq g(x)$, $x \in \mathbb{R}^+$. [6]



13-(MAT/11_HL_Summer_2015_Q5)-Functions-Roots

The functions f and g are defined by $f(x) = ax^2 + bx + c$, $x \in \mathbb{R}$ and $g(x) = p \sin x + qx + r$, $x \in \mathbb{R}$ where a, b, c, p, q, r are real constants.

(a) Given that f is an even function, show that $b=0$. [2]

(b) Given that g is an odd function, find the value of r . [2]

The function h is both odd and even, with domain $\mathbb{R}$.

(c) Find $h(x)$. [2]

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14-(MAT/11_HL_Summer_2015_Q6)-Functions-Roots, Integration

A function f is defined by $f(x) = \frac{3x-2}{2x-1}$, $x \in \mathbb{R}$, $x \neq \frac{1}{2}$.

(a) Find an expression for $f^{-1}(x)$. [4]

(b) Given that $f(x)$ can be written in the form $f(x) = A + \frac{B}{2x-1}$, find the values of the constants A and B . [2]

(c) Hence, write down $\int \frac{3x-2}{2x-1} dx$. [1]

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15-(MAT/11_HL_Summer_2015_Q7)-Functions-Roots

Let $p(x) = 2x^5 + x^4 - 26x^3 - 13x^2 + 72x + 36$, $x \in \mathbb{R}$.

(a) For the polynomial equation $p(x) = 0$, state

(i) the sum of the roots;

(ii) the product of the roots.

[3]

A new polynomial is defined by $q(x) = p(x + 4)$.

(b) Find the sum of the roots of the equation $q(x) = 0$.

[2]

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16-(MAT/11_HL_Summer_2015_Q9)-Functions-Roots

The functions f and g are defined by $f(x) = 2x + \frac{\pi}{5}$, $x \in \mathbb{R}$ and $g(x) = 3\sin x + 4$, $x \in \mathbb{R}$.

- (a) Show that $g \circ f(x) = 3\sin\left(2x + \frac{\pi}{5}\right) + 4$. [1]
- (b) Find the range of $g \circ f$. [2]
- (c) Given that $g \circ f\left(\frac{3\pi}{20}\right) = 7$, find the next value of x , greater than $\frac{3\pi}{20}$, for which $g \circ f(x) = 7$. [2]
- (d) The graph of $y = g \circ f(x)$ can be obtained by applying four transformations to the graph of $y = \sin x$. State what the four transformations represent geometrically and give the order in which they are applied. [4]

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17–(MAT/12_HL_Summer_2015_Q10)–Functions-Roots

The function f is defined by $f(x) = \frac{3x}{x-2}$, $x \in \mathbb{R}$, $x \neq 2$.

- (a) Sketch the graph of $y = f(x)$, indicating clearly any asymptotes and points of intersection with the x and y axes. [4]
- (b) Find an expression for $f^{-1}(x)$. [4]
- (c) Find all values of x for which $f(x) = f^{-1}(x)$. [3]
- (d) Solve the inequality $|f(x)| < \frac{3}{2}$. [4]
- (e) Solve the inequality $f(|x|) < \frac{3}{2}$. [2]



18-(MAT/12_HL_Summer_2015_Q11)-Functions-Roots,Differentiation,Integration

Consider the functions $f(x) = \tan x$, $0 \leq x < \frac{\pi}{2}$ and $g(x) = \frac{x+1}{x-1}$, $x \in \mathbb{R}$, $x \neq 1$.

(a) Find an expression for $g \circ f(x)$, stating its domain. [2]

(b) Hence show that $g \circ f(x) = \frac{\sin x + \cos x}{\sin x - \cos x}$. [2]

(c) Let $y = g \circ f(x)$, find an exact value for $\frac{dy}{dx}$ at the point on the graph of $y = g \circ f(x)$ where $x = \frac{\pi}{6}$, expressing your answer in the form $a + b\sqrt{3}$, $a, b \in \mathbb{Z}$. [6]

(d) Show that the area bounded by the graph of $y = g \circ f(x)$, the x -axis and the lines $x = 0$ and $x = \frac{\pi}{6}$ is $\ln(1 + \sqrt{3})$. [6]



19-(MAT/12_HL_Summer_2015_Q12)-Functions-Roots

The cubic equation $x^3 + px^2 + qx + c = 0$, has roots α, β, γ . By expanding $(x - \alpha)(x - \beta)(x - \gamma)$ show that

- (a) (i) $p = -(\alpha + \beta + \gamma)$;
(ii) $q = \alpha\beta + \beta\gamma + \gamma\alpha$;
(iii) $c = -\alpha\beta\gamma$. [3]

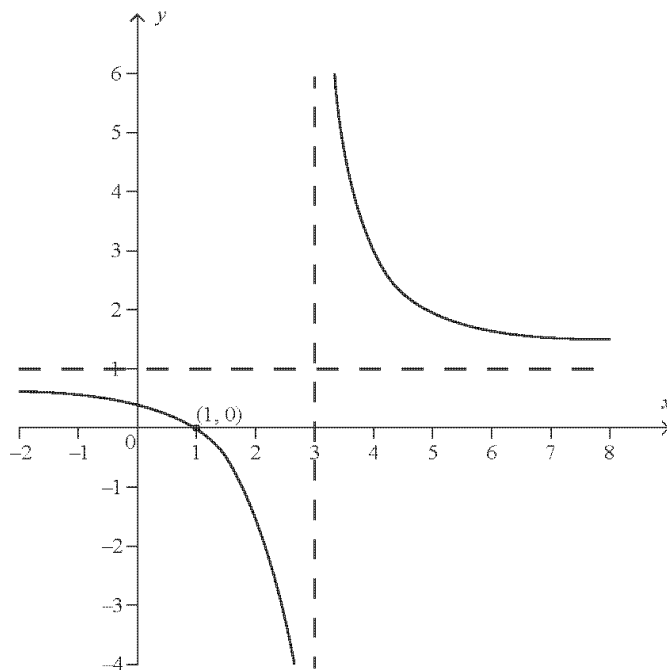
It is now given that $p = -6$ and $q = 18$ for parts (b) and (c) below.

- (b) (i) In the case that the three roots α, β, γ form an arithmetic sequence, show that one of the roots is 2.
(ii) Hence determine the value of c . [5]
- (c) In another case the three roots α, β, γ form a geometric sequence. Determine the value of c . [6]



20–(MAT/10_HL_Winter_2016_Q3)–Functions-Roots

A rational function is defined by $f(x) = a + \frac{b}{x - c}$ where the parameters $a, b, c \in \mathbb{Z}$ and $x \in \mathbb{R} \setminus \{c\}$. The following diagram represents the graph of $y = f(x)$.



Using the information on the graph,

- (a) state the value of a and the value of c ; [2]
- (b) find the value of b . [2]



21–(MAT/10_HL_Winter_2016_Q5)–*Functions-Roots*

[Maximum mark: 6]

The quadratic equation $x^2 - 2kx + (k - 1) = 0$ has roots α and β such that $\alpha^2 + \beta^2 = 4$.
Without solving the equation, find the possible values of the real number k .



22-(MAT/12_HL_Summer_2017_Q2)-Functions-Roots

The function f is defined by $f(x) = 2x^3 + 5$, $-2 \leq x \leq 2$.

- (a) Write down the range of f . [2]
- (b) Find an expression for $f^{-1}(x)$. [2]
- (c) Write down the domain and range of f^{-1} . [2]



23–(MAT/12_HL_Summer_2017_Q9)–Functions-Roots,Differentiation,Integration

Consider the function f defined by $f(x) = x^2 - a^2$, $x \in \mathbb{R}$ where a is a positive constant.

- (a) Showing any x and y intercepts, any maximum or minimum points and any asymptotes, sketch the following curves on separate axes.

(i) $y = f(x)$;

(ii) $y = \frac{1}{f(x)}$;

(iii) $y = \left| \frac{1}{f(x)} \right|$. [8]

- (b) Find $\int f(x) \cos x \, dx$. [5]

The function g is defined by $g(x) = x\sqrt{f(x)}$ for $|x| > a$.

- (c) By finding $g'(x)$ explain why g is an increasing function. [4]



24–(MAT/11_HL_Summer_2017_Q11)–Functions-Roots, Graphs, Integration

- (a) (i) Express $x^2 + 3x + 2$ in the form $(x + h)^2 + k$.
- (ii) Factorize $x^2 + 3x + 2$. [2]

Consider the function $f(x) = \frac{1}{x^2 + 3x + 2}$, $x \in \mathbb{R}$, $x \neq -2$, $x \neq -1$.

- (b) Sketch the graph of $f(x)$, indicating on it the equations of the asymptotes, the coordinates of the y -intercept and the local maximum. [5]
- (c) Show that $\frac{1}{x+1} - \frac{1}{x+2} = \frac{1}{x^2 + 3x + 2}$. [1]
- (d) Hence find the value of p if $\int_0^1 f(x) dx = \ln(p)$. [4]
- (e) Sketch the graph of $y = f(|x|)$. [2]
- (f) Determine the area of the region enclosed between the graph of $y = f(|x|)$, the x -axis and the lines with equations $x = -1$ and $x = 1$. [3]



25–(MAT/11_HL_Summer_2017_Q12)–Functions-Roots, Graphs

Consider the polynomial $P(z) = z^5 - 10z^2 + 15z - 6$, $z \in \mathbb{C}$.

(a) Write down the sum and the product of the roots of $P(z) = 0$. [2]

(b) Show that $(z - 1)$ is a factor of $P(z)$. [2]

The polynomial can be written in the form $P(z) = (z - 1)^3(z^2 + bz + c)$.

(c) Find the value of b and the value of c . [5]

(d) Hence find the complex roots of $P(z) = 0$. [3]

Consider the function $q(x) = x^5 - 10x^2 + 15x - 6$, $x \in \mathbb{R}$.

(e) (i) Show that the graph of $y = q(x)$ is concave up for $x > 1$.

(ii) Sketch the graph of $y = q(x)$ showing clearly any intercepts with the axes. [6]



26-(MAT/12_HL_Summer_2018_Q10)-Functions-Roots

The function f is defined by $f(x) = \frac{ax+b}{cx+d}$, for $x \in \mathbb{R}$, $x \neq -\frac{d}{c}$.

- (a) Find the inverse function f^{-1} , stating its domain. [5]

The function g is defined by $g(x) = \frac{2x-3}{x-2}$, $x \in \mathbb{R}$, $x \neq 2$.

- (b) (i) Express $g(x)$ in the form $A + \frac{B}{x-2}$ where A, B are constants. [5]
(ii) Sketch the graph of $y = g(x)$. State the equations of any asymptotes and the coordinates of any intercepts with the axes.

The function h is defined by $h(x) = \sqrt{x}$, for $x \geq 0$.

- (c) State the domain and range of $h \circ g$. [4]



27–(MAT/12_HL_Summer_2019_Q3)–Functions-Roots

[Maximum mark: 5]

Consider the function $f(x) = x^4 - 6x^2 - 2x + 4$, $x \in \mathbb{R}$.

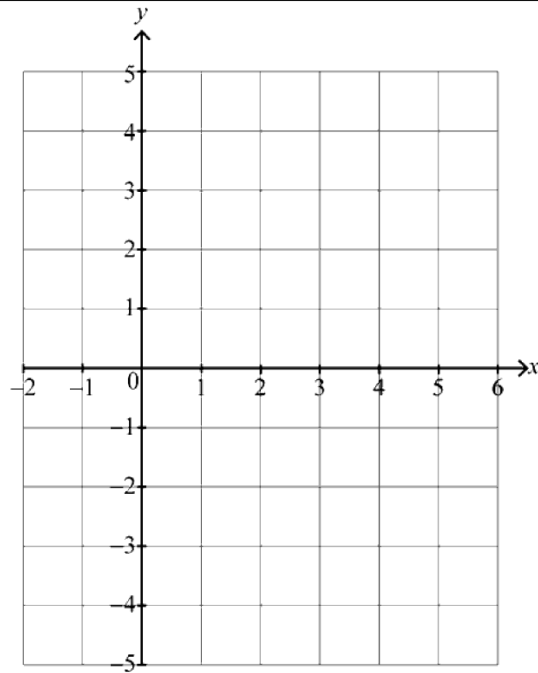
The graph of f is translated two units to the left to form the function $g(x)$.

Express $g(x)$ in the form $ax^4 + bx^3 + cx^2 + dx + e$ where $a, b, c, d, e \in \mathbb{Z}$.



28-(MAT/12_HL_Summer_2019_Q5)-Graphs, Functions-Roots

- (a) Sketch the graph of $y = \frac{x-4}{2x-5}$, stating the equations of any asymptotes and the coordinates of any points of intersection with the axes. [5]



- (b) Consider the function $f: x \rightarrow \sqrt{\frac{x-4}{2x-5}}$.

Write down

- (i) the largest possible domain of f ;
(ii) the corresponding range of f .

[3]



29–(MathAA/12_HL_Summer_2021_Q1)–Algebra-functions-Roots

[Maximum mark: 4]

Consider two consecutive positive integers, n and $n + 1$.

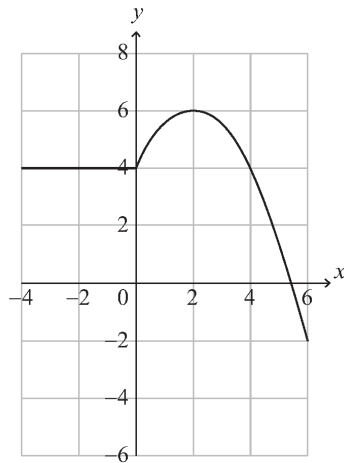
Show that the difference of their squares is equal to the sum of the two integers.



30-(MathAA/11_HL_Summer_2021_Q1)-Algebra-functions-Roots

[Maximum mark: 5]

The graph of $y = f(x)$ for $-4 \leq x \leq 6$ is shown in the following diagram.



(a) Write down the value of

(i) $f(2)$;

(ii) $(f \circ f)(2)$.

[2]

(b) Let $g(x) = \frac{1}{2}f(x) + 1$ for $-4 \leq x \leq 6$. On the axes above, sketch the graph of g .

[3]



31–(MathAA/12_HL_Summer_2021_Q7)–Algebra-functions-Roots

[Maximum mark: 5]

The cubic equation $x^3 - kx^2 + 3k = 0$ where $k > 0$ has roots α , β and $\alpha + \beta$.

Given that $\alpha\beta = \frac{k^2}{4}$, find the value of k .



32–(MathAA/10_HL_Winter_2021_Q2)–Algebra-functions-Roots, Graphs

[Maximum mark: 9]

The function f is defined by $f(x) = \frac{2x+4}{3-x}$, where $x \in \mathbb{R}$, $x \neq 3$.

(a) Write down the equation of

(i) the vertical asymptote of the graph of f ;(ii) the horizontal asymptote of the graph of f .

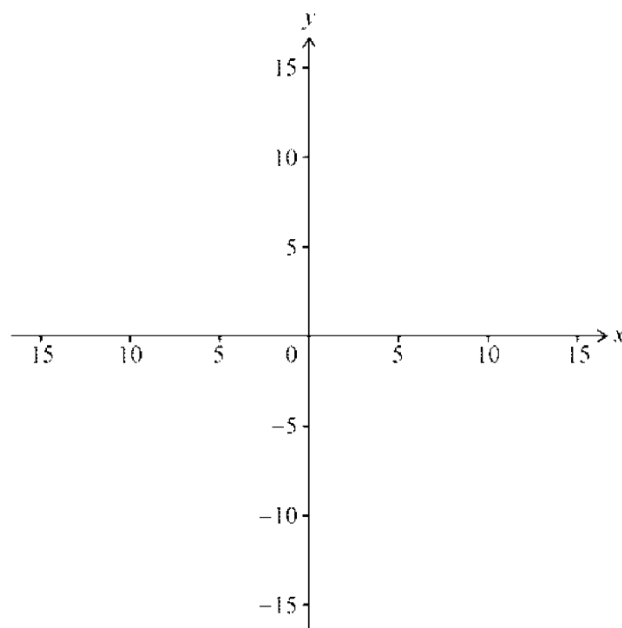
[2]

(b) Find the coordinates where the graph of f crosses(i) the x -axis;(ii) the y -axis.

[2]

(c) Sketch the graph of f on the axes below.

[1]



The function g is defined by $g(x) = \frac{ax+4}{3-x}$, where $x \in \mathbb{R}$, $x \neq 3$ and $a \in \mathbb{R}$.

(d) Given that $g(x) = g^{-1}(x)$, determine the value of a .

[4]



33-(MathAA/12_HL_Summer_2022_Q11)-Algebra-functions-Roots

[Maximum mark: 20]

A function f is defined by $f(x) = \frac{1}{x^2 - 2x - 3}$, where $x \in \mathbb{R}$, $x \neq -1$, $x \neq 3$.

- (a) Sketch the curve $y = f(x)$, clearly indicating any asymptotes with their equations. State the coordinates of any local maximum or minimum points and any points of intersection with the coordinate axes. [6]

A function g is defined by $g(x) = \frac{1}{x^2 - 2x - 3}$, where $x \in \mathbb{R}$, $x > 3$.

- (b) The inverse of g is g^{-1} .

(i) Show that $g^{-1}(x) = 1 + \frac{\sqrt{4x^2 + x}}{x}$.

- (ii) State the domain of g^{-1} . [7]

A function h is defined by $h(x) = \arctan \frac{x}{2}$, where $x \in \mathbb{R}$.

- (c) Given that $(h \circ g)(a) = \frac{\pi}{4}$, find the value of a .

Give your answer in the form $p + \frac{q}{2}\sqrt{r}$, where $p, q, r \in \mathbb{Z}^+$. [7]



34--(MathAA/12_HL_Summer_2022_Q9)--Algebra-functions-Roots

[Maximum mark: 5]

Prove by contradiction that the equation $2x^3 + 6x + 1 = 0$ has no integer roots.

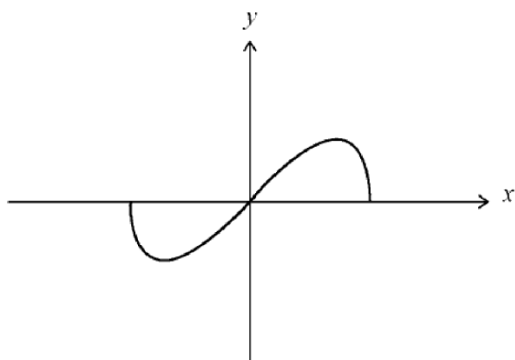


35-(MathAA/12_HL_Summer_2022_Q6)-Algebra-functions-Roots

[Maximum mark: 8]

A function f is defined by $f(x) = x\sqrt{1-x^2}$ where $-1 \leq x \leq 1$.

The graph of $y = f(x)$ is shown below.



(a) Show that f is an odd function.

[2]

The range of f is $a \leq y \leq b$, where $a, b \in \mathbb{R}$.

(b) Find the value of a and the value of b .

[6]



36–(MathAA/12_HL_Summer_2022_Q2)–Algebra-functions-Roots

[Maximum mark: 6]

Consider any three consecutive integers, $n - 1$, n and $n + 1$.

- (a) Prove that the sum of these three integers is always divisible by 3. [2]
- (b) Prove that the sum of the squares of these three integers is never divisible by 3. [4]



37–(MathAA/11_HL_Summer_2022_Q8)–Algebra-functions-Roots

[Maximum mark: 6]

Consider integers a and b such that $a^2 + b^2$ is exactly divisible by 4. Prove by contradiction that a and b cannot both be odd.



38-(MathAA/11_HL_Summer_2022_Q3)-Algebra-functions-Roots

[Maximum mark: 7]

Consider the functions $f(x) = \sqrt{3} \sin x + \cos x$ where $0 \leq x \leq \pi$ and $g(x) = 2x$ where $x \in \mathbb{R}$.

- (a) Find $(f \circ g)(x)$. [2]
- (b) Solve the equation $(f \circ g)(x) = 2 \cos 2x$ where $0 \leq x \leq \pi$. [5]



39—(MathAA/10_HL_Winter_2022_Q8)—Trigonometry, Algebra-functions-Roots

[Maximum mark: 5]

Let $f(x) = \cos(x - k)$, where $0 \leq x \leq a$ and $a, k \in \mathbb{R}$.

- (a) Consider the case where $k = \frac{\pi}{2}$.

By sketching a suitable graph, or otherwise, find the largest value of a for which the inverse function f^{-1} exists.

[2]

- (b) Find the largest value of a for which the inverse function f^{-1} exists in the case where $k = \pi$.

[1]

- (c) Find the largest value of a for which the inverse function f^{-1} exists in the case where $\pi < k < 2\pi$. Give your answer in terms of k .

[2]



40—(MathAA/10_HL_Winter_2022_Q3)—Algebra-functions-Roots

[Maximum mark: 5]

Let a be a constant, where $a > 1$.

(a) Show that $a^2 + \left(\frac{a^2 - 1}{2}\right)^2 = \left(\frac{a^2 + 1}{2}\right)^2$. [3]

Consider a right-angled triangle with sides of length a , $\left(\frac{a^2 - 1}{2}\right)$ and $\left(\frac{a^2 + 1}{2}\right)$.

(b) Find an expression for the area of the triangle in terms of a . [2]



1-(MAT/11_HL_Summer_2013_Q12)-Quadratics, Graphs, Integration

(a) Express $4x^2 - 4x + 5$ in the form $a(x - h)^2 + k$ where $a, h, k \in \mathbb{Q}$. [2 marks]

(b) The graph of $y = x^2$ is transformed onto the graph of $y = 4x^2 - 4x + 5$. Describe a sequence of transformations that does this, making the order of transformations clear. [3 marks]

The function f is defined by $f(x) = \frac{1}{4x^2 - 4x + 5}$.

(c) Sketch the graph of $y = f(x)$. [2 marks]

(d) Find the range of f . [2 marks]

(e) By using a suitable substitution show that $\int f(x) dx = \frac{1}{4} \int \frac{1}{u^2 + 1} du$. [3 marks]

(f) Prove that $\int_1^{3.5} \frac{1}{4x^2 - 4x + 5} dx = \frac{\pi}{16}$. [7 marks]

